Heat cramps

Heat cramps are painful muscle cramps caused by losing too much salt through sweating. They are usually the result of heavy exercise or physical work in a hot environment. Heat cramps usually occur in the muscles that have been used the most strenuously, such as those in the legs and abdomen. The cramps typically occur late in a workday or after the muscles have cooled (e.g., during a shower after work).

It is important to distinguish heat cramps from the more common cramps that occur during strenuous work. Common cramps will be cured with rest and massage. Heat cramps are cured only after the lost salt is replaced.

Signs and symptoms

- Muscular pain or spasms
- Excessive sweating

Treatment

- Move the worker to a cooler environment. If possible, lay the worker down and remove or loosen tight-fitting clothing.
- Cool the worker by sponging with cool water and fanning. Don't cool the worker too much. If the worker begins to shiver, stop cooling.
- If the worker is fully alert and not nauseated, provide oral fluids. The best fluids are juice, non-caffeinated soft drinks, commercially available oral rehydrating solutions (sport drinks), or a solution of salt water (1 teaspoon of salt in 0.5 litres of water). Alcoholic and caffeinated beverages are not recommended.

Continuing to work under conditions of heat stress can lead to heat exhaustion, a more serious illness. If a worker has cramps and other symptoms (e.g., fatigue, weakness, headache, gastrointestinal illness, or changes in mental state), then they may already be suffering from heat exhaustion or heat stroke. These conditions are medical emergencies that need to be treated as such.

Heat exhaustion

Heat exhaustion is caused by losing both water and salt through sweating. It can happen during prolonged periods of exertion, when fluid replacement has not matched losses. It is more serious than heat cramps. The worker will have a number of other signs and symptoms.

Signs and symptoms

- Shallow respiration
- Increased respiratory rate
- Weak, rapid pulse
- Cool, pale, clammy skin

- Sweating
- Weakness, fatigue, dizziness
- Headache and nausea
- Fainting
- Muscle cramps

Signs and symptoms are the same as for mild shock. The presence of sweating is an important finding. It is often the only way to differentiate heat exhaustion from life-threatening heat stroke. If untreated, heat exhaustion may progress to heat stroke. Workers suffering from heat exhaustion should be transported to medical aid.

Treatment

- Move the worker to a cooler environment. If possible, lay them down and remove or loosen tight-fitting clothing.
- Cool the worker by sponging with cool water and fanning. Don't cool the worker too much. If the worker begins to shiver, stop cooling.
- If the worker is fully alert and not nauseated, provide oral fluids. The best fluids are juice, non-caffeinated soft drinks, commercially available oral rehydrating solutions (sport drinks), or a solution of salt water (1 teaspoon of salt in 0.5 litres of water). Alcoholic and caffeinated beverages are not recommended.

In most cases, the worker's symptoms will improve dramatically within 30 minutes. They should still be transported to medical aid.

Heat stroke

Heat stroke occurs when the body's mechanisms for heat dissipation are overwhelmed and fail. Heat stroke is a life-threatening condition in which the body's core temperature rises above 41°C. At core body temperatures over 41°C, sweating stops and the body is unable to get rid of heat, causing body temperature to continue to rise. The person's mental functions may become disturbed. Without immediate first aid, heat stroke can result in loss of consciousness, permanent brain damage, and death. Notify a first aid attendant, and arrange for transportation to medical aid immediately.

Note

Heat stroke is a medical emergency. Notify a first aid attendant, call 911, and arrange for immediate transportation to medical aid.

Heat stroke can occur quickly and without warning. It may be preceded by heat exhaustion (see page 20).

Signs and symptoms

- Hot, dry, flushed skin (or profuse sweating in some cases)*
- Agitation, confusion
- Decreased level of consciousness
- Headache
- Nausea and vomiting
- Seizures
- Increased respiratory rate
- Irregular pulse rate
- Shock
- Cardiac arrest

* Heat stroke is generally identified by the presence of hot, dry, flushed skin, without evidence of sweating. However, profuse sweating may be present during heat stroke that occurs because of strenuous physical activity.

Treatment

- Maintain airway, breathing, and circulation as required.
- Move the worker to the coolest place available.
- Lay the worker down on their back unless the worker is actively vomiting or having a seizure. If they are, place them in the three-quarter prone or on-the-side position.
- Remove all outer clothing and apply cold water to the worker by either dousing or applying wet cool sheets. Spraying or sponging the entire body with cold water is also effective. Fanning will help.
- If the worker is fully alert and not nauseated, provide oral fluids. The best fluids are juice, non-caffeinated soft drinks, commercially available oral rehydrating solutions (sport drinks), or a solution of salt water (1 teaspoon of salt in 500 mL of water). Alcoholic and caffeinated beverages are not recommended.
- Transport the worker to medical aid and continue to cool during transport.

BK30 Printed in Canada R05/26